A perfect split from two support bounds

Human Compression project: a component of the L019 recovery

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle$ be a transitive string action by involutions on n points. Assume that its noncommutation diagram is connected, every maximal omission is intransitive, and $2r > n + 2$. An eligible edge of colour i joins different orbits of the actual maximal omission G_i. If there is exactly one such edge, deleting it from the full permutation graph gives the two G_i-orbits. Call this cut perfect if, after choosing names A, B for these orbits, $G_{<i}$ fixes B pointwise and $G_{>i}$ fixes A pointwise. The generator ρ_i may still have internal edges in both sides.

We use three complete earlier arguments: paired orbit cuts, including the bound $2r \le n$ when every colour has two eligible edges; the orbit-budget and head-support theorems in their stated parity-free form; and the ambient interval inequality. The last says that an intransitive subgroup generated by a nonempty contiguous interval I, each of whose colours has two ambient eligible edges, has support U satisfying $2|I| \le |U| - c$, where $c \ge 1$ is its number of nontrivial orbits. No eligibility relative to a smaller subgroup is substituted for this hypothesis.

Lemma 1 (Two separated unique cuts)**.** *Suppose unique cuts at $i < j$ have edges in series. Write their three regions as A, M, D, so the i-omission orbits are A and $B = M \cup D$, while the j-omission orbits are $C = A \cup M$ and D. Suppose $j \ge i + 2$, $G_{<i}$ fixes B, and $G_{>j}$ fixes C. If $h > i$ acts on A and $g < j$ acts on D, then*

$$n \ge 2r + h - g.$$

The assertion also allows $i = 0$ or $j = r - 1$.

Proof. Apply the outside-side bound at i to get $|A| \ge i+h+1$. Apply it with all indices reversed at j: the crossing index is $r-1-j$ and the active index is $r-1-g$. Thus $|D| \ge 2r - j - g - 1$.

Let $b \in M$ be the endpoint of the i-edge towards j, and $c \in M$ the endpoint of the j-edge towards i. A simple path from b to c in M avoids colour i's unique crossing and stays in B. The no-gap argument at i applies to any such path to a point moved by ρ_j: it must contain every index $i + 1, \dots, j - 1$. Equivalently, if some such index k were missing, commute the letters of index greater than k to the beginning. They all fix b, while ρ_j fixes b and commutes with the remaining lower-index word. It would fix c, which it does not. Hence the path has at least $j - i - 1$ edges and $|M| \ge j - i$. Summing the three disjoint region bounds proves the assertion. □

Theorem 2. *Under the standing hypotheses there is a perfect split.*

Proof. The paired-cut bound ensures a unique eligible edge for at least one colour. Suppose, for a contradiction, that no unique cut is perfect. The orbit-budget theorem then provides a shared mark on one side of some cut; reverse the entire index order if necessary to put it to the right. In this fixed order choose a largest cut index i having some shared mark of index greater than i. The primitive interval side furnished by the orbit-budget argument must lie on the right, since every shared mark acts on that side. Thus name the sides so that $G_{<i}$ fixes B and $G_{>i}$ is transitive on B. Both sides have size at least two: an interior singleton contradicts the singleton budget, while an endpoint singleton is already a perfect split.

Let $h > i$ be the largest shared index. The head theorem gives $h < r-1$ and, with $X = \mathrm{supp}(G_{>h})$,

$$X \subseteq B \setminus \{b\}, \qquad 2h \le n - |X| - 1. \tag{1}$$

At $i = 0$ we choose the names of the sides according to the final tail, as expressly allowed by that theorem; the condition on $G_{<i}$ remains automatic. If every colour greater than h had two ambient eligible edges, the interval theorem, applied to the tail fixing A, would give $2(r-1-h) \le |X|$. Together with (1) this would imply $2r \le n+1$, a contradiction.

Choose the least $j > h$ having a unique eligible edge. Since $G_{>h}$ is supported in B, this edge lies in B. The two unique edges are graph bridges, so deleting both partitions the connected full graph into precisely three regions A, M, D, in that order. Thus the j-orbits are $C = A \cup M$ and D. The region M contains the two endpoints facing one another, and $j \ge i+2$. By maximality of i, no mark greater than j is shared at this new cut. Since this cut is not perfect, it has a shared mark less than j. Let g be its first shared index. Again $|C|, |D| \ge 2$.

We claim that $G_{>j}$ fixes C, and that the reversed head theorem has its final tail in C. If $j < r-1$, the primitive interval at j lies to the left, since there is a shared mark to the left and none to the right. If this primitive side were D, then $G_{>j}$ would fix D. Apply the last-mark bound with indices reversed, with first side C and primitive side D. Its crossing index $r-1-j$ is positive. The original mark ρ_0 acts nontrivially on C: for $i > 0$ it has its nonempty support in $A \subset C$, and for $i = 0$ it contains the i-bridge, which lies wholly in C. The last-mark bound would give $n \ge 2r$, impossible. Therefore the primitive side is C, proving that $G_{>j}$ fixes it. The reversed head theorem now places its final tail in C.

If $j = r-1$, $G_{>j}$ is empty and fixes C automatically. The reversed last-mark bound excludes $g = 0$, since that would make the original mark ρ_0 shared at this endpoint cut. The final reversed tail acts on a single side, and it contains ρ_0, which acts nontrivially on C as just shown. Its side is therefore C. In both cases, with $Y = \mathrm{supp}(G_{<g})$, the reversed head bound is exactly

$$Y \subseteq C, \qquad 2(r-1-g) \le n - |Y| - 1. \tag{2}$$

The reversed index is $r-1-g$, not $r-g$.

Lemma 1 applies, because h acts on A and g on D. It gives $n \ge 2r + h - g$. Thus $g \le h$ is impossible, and $g > h$. The two groups defining X, Y together contain every mark. Since G is transitive on more than one point, no point is fixed by all its generators. Consequently

$$X \cup Y = \Omega. \tag{3}$$

Every colour strictly between h and g has two ambient eligible edges, by the minimal choice of j and $g < j$. Its subgroup has support $V \subseteq X \cap Y$ and fixes both i-omission sides setwise, so is intransitive. If this interval is nonempty, the ambient interval bound gives $2(g-h-1) \le |V| \le |X \cap Y|$. If it is empty, the same weak inequality is immediate. Summing it with (1) and (2), and then using (3), gives

$$\begin{aligned} 2r &= 2h + 2(g-h-1) + 2(r-1-g) + 4 \\ &\le (n - |X| - 1) + |X \cap Y| + (n - |Y| - 1) + 4 \\ &= n + 2, \end{aligned}$$

contradicting the standing strict inequality. This proves the theorem. □

Relation to L019 and attribution. Cameron–Fernandes–Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2, establish perfect-split existence in Section 5. The fracture strategy, unique crossings and support overlap remain attributed to that source and the earlier alternating-group work. The present assembly uses the complete acquired ambient interval proof and the newly reworked actual-orbit support bounds. Its separated-cut argument

retains all weak equality cases; the strict inequality $2r > n + 2$ is what excludes them. No parity-specific graph classification or quotient IP is silently imported. The global middle-colour exchange in the acquired paired-cut theory retains its separately disclosed root contribution and distinct review.

For a high-rank symmetric C-marking, the accepted initial fracture theorem supplies the standing omission hypothesis in its stated degree range; its small-degree qualification must still be retained when assembling the principal source. This theorem alone does not establish a pure end mark, the degree/rank bijection, or the six enumerative constants. No whole-source credit, new conjecture, Lean declaration or compression measurement is claimed.